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Pharmaceutical polymer-based hydrogels for 3D bioprinted drug delivery and tissue engineering applications

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Abstract

Introduction 3D bioprinting enables the layer-by-layer fabrication of living tissue constructs and supports patient-specific customization. This technology holds strong promises for regenerative medicine and drug discovery. However, its broader translation remains limited by material variability, safety considerations, cost constraints and regulatory requirements.

Background This study investigates the use of safe, affordable and regulatory-compliant pharmaceutical polymers as biomaterials for 3D bioprinting. It specifically focuses on hydrogels formulated from Starch 1500®, maltodextrin and sodium alginate. The objective is to assess their potential applications in skin tissue engineering and oral drug delivery through semisolid extrusion-based 3D bioprinting techniques.

Results Ionic crosslinking of the hydrogel was confirmed by FTIR analysis. The hydrogel exhibited a viscosity of 1.56×10^6 mPa·s, supporting semisolid extrusion bioprinting and excellent printability under ambient conditions. It enabled the fabrication of multilayer scaffolds with uniform filaments, well-defined square pore geometry and good shape fidelity. Rheological analysis showed shear-thinning behavior under applied stress, 87% thixotropic recovery and predominantly solid-like behavior at rest. Cast films showed a tensile strength of 33.9 MPa with limited extensibility, whereas lyophilized scaffolds exhibited high porosity and an average pore size of 39.2 μ m. The 3D-printed scaffolds swollen upto 72% within 24 h and showed the onset of degradation after 2 weeks. Biological evaluation confirmed non-cytotoxicity, with more than 70% cell viability in skin-relevant L929 and HaCaT cells and good hemocompatibility, indicated by 5.0% hemolysis. Confocal microscopy further showed cell growth on the crosslinked hydrogel, supporting their potential for skin tissue engineering. Glimepiride-loaded bioinks were successfully formulated into chewable tablets for drug delivery, which exhibited acceptable physical properties. The tablets demonstrated excellent content uniformity (100.4%), while dissolution results showed sustained-release profiles over a time period of four hours.

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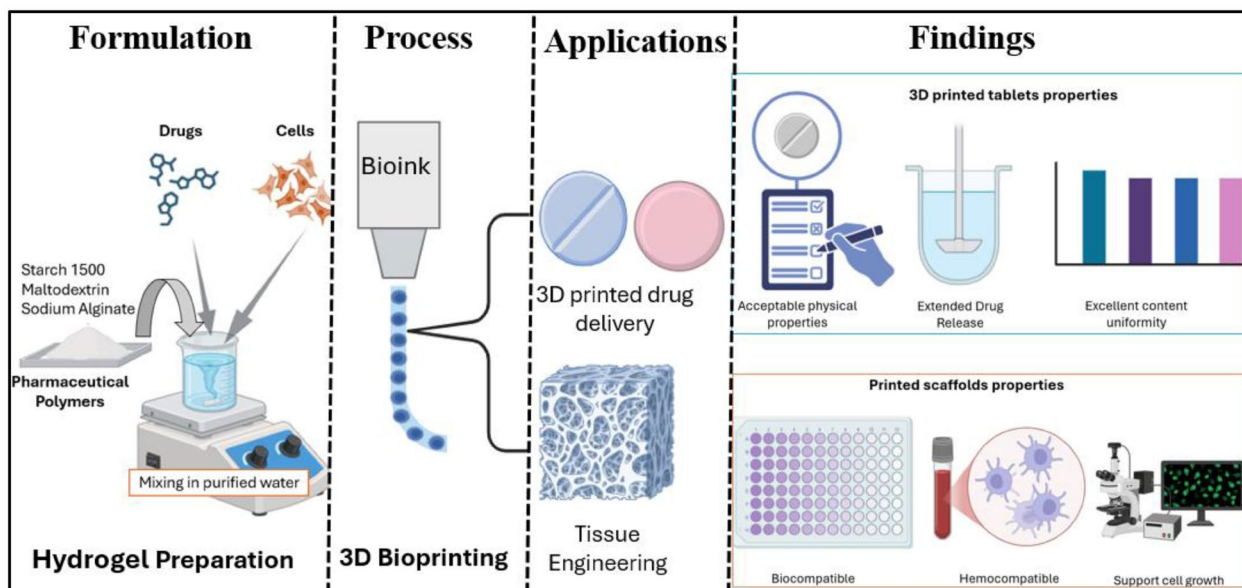


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Conclusions Our research indicates that pharmaceutical-grade polymer-based hydrogels are promising candidates for skin tissue engineering and drug delivery through 3D bioprinting. The findings of this study underscore the potential of these formulations to advance bioprinting technologies and related applications.

Keywords Starch 1500®, Pharmaceutical polymers, Hydrogels, Semisolid extrusion, 3D-printed tablets, Tissue engineering

Graphical Abstract



Introduction

Three-dimensional (3D) printing is an innovative technology that constructs objects layer by layer. In healthcare, it enables the customization of implants, prosthetics and medical devices for individual patients. Key benefits include enhanced personalization, precision, reduced production times and cost-effectiveness, all of which contribute to improved patient outcomes [1].

3D bioprinting is an advanced form of 3D printing that utilizes bioinks to create complex biological structures for applications in tissue engineering and regenerative medicine [2]. The development of organs for transplantation [3], accurate disease models for medical research and drug testing [4, 5], the production of sustainable artificial meat [6] and the design of advanced 3D-printed drug delivery systems that enhance patient compliance and therapeutic efficacy [7, 8]. A significant advantage of 3D bioprinting is its ability to fabricate personalized scaffolds from the patient's own cells, thereby reducing the risk of immune rejection [9, 10].

3D-printed tablets support rapid prototyping and on-demand manufacturing, thereby accelerating product development and reducing material waste. They also enable customization of shape, size, excipients (including colorants and flavours), dose, and drug-release profile to meet individual patients' needs better. By incorporating

personalized medicine features, 3D-printed tablets can enhance patient compliance and improve therapeutic outcomes [11]. In 2015, the US-FDA (United States Food and Drug Administration) approved the first 3D-printed SPRITAM® (levetiracetam) tablets [12]. Since then, many researchers and pharmaceutical companies have been exploring the use of 3D printed medicine. As a result, the market for 3D printed drugs is expected to grow rapidly, reaching \$24.3 billion by 2030 [13].

Despite significant progress, researchers in 3D bioprinting still face challenges with biomaterials, including regulatory compliance, immunogenicity, batch-to-batch variability, limited raw material availability, and mechanical stability and affordability [14]. Pharmaceutical polymers have the potential to address many of these limitations. They are animal-free and FDA-approved, offer consistent quality and safety, and are widely available at competitive cost [15–17].

This study evaluates pharmaceutical polymer-based hydrogels as bioinks for semisolid extrusion-based 3D bioprinting, with potential applications in skin tissue engineering and drug delivery.

Table 1 Hydrogel formulation for tissue engineering

Ingredients	Properties	% w/w	Batch quantity, grams
Starch 1500®	Viscosity modifier	3.00	3.00
Maltodextrin	Printability aid	3.00	3.00
Sodium Alginate	Gelling Agent	9.00	9.00
Purified water, in mL	Media	85.00	85.00
	Total	100	100
Calcium chloride solution (5% w/v)	Crosslinker	9:1 ratio of hydrogel (9 mL): crosslinker (1 mL)	

Materials and methods

Materials and preparation of hydrogel for tissue engineering applications

The hydrogel was prepared using Starch 1500® (partially pregelatinized maize starch; Colorcon, USA), maltodextrin (Glucidex 12D; Roquette, France), and sodium alginate (Protanal 8133; DuPont Nutrition, Scotland). Starch 1500® was incorporated to enhance viscosity and support cell–material interactions, whereas maltodextrin was used to modulate flow behavior and improve extrusion performance. Sodium alginate was included as the gelling polymer because of its ability to undergo ionic crosslinking with calcium ions, thereby contributing to gel formation and structural integrity. Purified water obtained from an ELGA PURELAB Option DV35 system (Metrohm, India) was used as the dispersion medium.

For hydrogel preparation, 15 g of the polymer mixture was gradually dispersed in 85 mL of purified water under continuous magnetic stirring to obtain a 15% w/v formulation. Stirring continued for 90 min until a smooth, homogeneous hydrogel formed, followed by autoclaving at 121 °C for 15 min to sterilize. After cooling to room temperature, the hydrogel was crosslinked with sterile calcium chloride solution (5% w/v) at a hydrogel-to-crosslinker ratio of 9:1 using a syringe fitted with a female luer-lock adaptor. The detailed formulation composition is provided in Table 1.

Physicochemical properties of hydrogel and 3D bioprinted scaffolds

Printability of hydrogel Printability was assessed using a crosslinked hydrogel containing a food-grade dye to aid visual observation during printing. Printing was performed using pneumatic, semisolid extrusion-based 3D bioprinting with a Trivima Basic bioprinter (Next Big Innovation Labs, Bengaluru, India) fitted with a 10 mL Luer-Lock syringe and a 22-gauge tapered tip. Square-grid and spiral scaffolds of different dimensions were fabricated to evaluate shape fidelity, filament uniformity, pore geometry, and multilayer printing capability. Filament diameter and pore geometry were measured from

Table 2 Printing setup and process parameters for hydrogel in Trivima Basic bioprinter

3-D Bioprinter	Trivima Basic, NBIL, Bengaluru, India
Printhead Z movement mechanism	Pneumatic pressure
3D printing control software	Niyantranam
Printing base	35 mm Petri plate
Syringe	10 ml luer lock
Dispensing needle	Tapered tips 22 Gauge, luer lock
Dimensions (Xx Yx Z), mm	Single-layered- 10×10×0.4 mm Five layered- 30×30×2.0 mm
Shape	Square and spiral
Number of layers	1–5
Speed, mm/min	250
Extruder temperature, °C	25
Bed temperature, °C	25
Layer height, mm	0.4
Pressure, Bar	1.5

images of the printed scaffolds using the ruler/measurement function in Photopea (online image analysis software). Additionally, the printed constructs were visually examined for smooth extrusion, continuous filament deposition, preservation of the intended architecture, and evidence of nozzle clogging or structural collapse. The printing setup and process parameters used for this evaluation are summarized in Table 2.

Viscosity testing of hydrogel Viscosity measurements of the crosslinked hydrogel was performed using a Brookfield Ametek DV2T-RV viscometer to evaluate its suitability for bioprinting. Before analysis, the hydrogel was equilibrated at room temperature, and measurements were conducted in triplicate at 22 ± 2 °C. The final viscosity was reported as the mean $\pm$ standard deviation of three measurements.

Rheological testing of hydrogel Rheological characterization of the crosslinked hydrogel was conducted at 28 °C using approximately one spatulaful of sample on an Anton Paar MCR 302 rheometer fitted with a rough cone–plate geometry (40 mm diameter, 1° cone angle, and 5.79 μ m surface roughness). The hydrogel was evaluated for shear-thinning behaviour, thixotropy, viscoelasticity, and yield stress [τ_y], and flow-point [τ_f]. Before each test, the sample was preconditioned by applying an oscillatory stress of 300 Pa at 10 rad/s for 10 min, followed by a 30-minute rest under 0.1% oscillatory strain at 1 rad/s to improve repeatability.

Shear-thinning, yield stress and flow point were assessed by a shear rate ramp test in which the shear rate was increased from 0.01 to 1000 s⁻¹. Thixotropy was evaluated using a viscosity recovery test; the sample was subjected to 0.1 s⁻¹ for 5 min, then 100 s⁻¹ for 10 min, and

finally 0.1 s^{-1} for 15 min. Viscosity recovery was calculated as the ratio of recovered viscosity to initial viscosity. Viscoelasticity was examined using oscillatory testing; a strain-amplitude sweep from 0.01% to 10,000% at 1 rad/s, and a frequency sweep from 100 to 0.01 rad/s at a constant strain of 0.5%.

% Swelling and in vitro degradation rate of 3D bio-printed scaffolds To evaluate the swelling and degradation rates of 3D printed scaffolds, we printed six eight-layered scaffolds in a square grid of $10 \text{ mm} \times 10 \text{ mm}$, making up six scaffolds each for the swelling and in vitro biodegradation study.

After printing, the scaffolds were post-crosslinked and weighed to get the initial weight. They were then placed in 2 mL of DMEM (Dulbecco's Modified Eagle Medium) media at room temperature. To measure swelling, we recorded the wet weight (swollen) of the scaffolds at 15 min, 30 min, 1 h, 5 h, and 24 h. The swelling percentage was calculated using Eq. (1):

$$\% \text{ Swelling} = \left(\frac{\text{Wet weight} - \text{Initial weight}}{\text{Initial weight}} \right) \times 100 \quad (1)$$

The in vitro degradation of 3D printed scaffolds was evaluated by calculating the percentage of mass loss over time. To simulate physiological conditions, scaffold samples were incubated at 37°C , and the incubation media were replaced every other day. Throughout the study, scaffolds swelled until day fourteen, after which a decrease in weight indicated the onset of degradation.

At predetermined time points (0, 7, 14, 17, 21, 28, and 35 days), scaffolds were removed from the incubation medium, gently blotted to remove excess medium, and weighed to obtain the final weight. Scaffold degradation was quantified as percentage mass loss using Eq. (2), providing a time-dependent assessment of degradation throughout the incubation period.

$$\% \text{ Mass loss} = \left(\frac{\text{Initial weight} - \text{Final weight}}{\text{Initial weight}} \right) \times 100 \quad (2)$$

Porosity of the scaffolds through FESEM To investigate the surface morphology of the 3D-printed scaffolds, the printed constructs were freeze-dried using a Labconco lyophilizer. Following lyophilization, the scaffolds were treated with a 2.5% glutaraldehyde solution and stored overnight at 4°C to ensure thorough fixation. After fixation, the scaffolds were washed with 1x phosphate-buffered saline (PBS) to remove residual fixative. A sequential dehydration protocol was then employed, in which the scaffolds were immersed in ethanol solutions of increasing concentrations (70%, 80%, 90%, and finally 100%) for 10 min at each step. After dehydration, the sam-

ples underwent critical point drying in a LEICA EM CPD 300 for critical point drying (CPD) for 15 cycles to prevent structural collapse. Before imaging, a 4 nm layer of gold-palladium (Au-Pd) was sputter-coated onto the scaffold surfaces to preserve their morphology and enhance imaging quality.

The lyophilized 3D-printed scaffolds were analyzed using a Quanta FEG 250 field-emission scanning electron microscope (FESEM) at magnifications of 1000x and 2500x. Pore sizes at a total of fifteen pores and percentage porosity were quantified in ImageJ from images captured at these magnifications. The mean pore diameter and percentage porosity were recorded, along with their standard deviations.

Chemical analysis of the scaffolds through FTIR Fourier Transform Infrared spectroscopy (FTIR) analysis was performed using a Bruker Alpha II Platinum spectrophotometer with OPUS software. Transmittance spectra were recorded over $400\text{--}4000 \text{ cm}^{-1}$. Spectra of the uncrosslinked and crosslinked hydrogels were compared to evaluate chemical changes caused by crosslinking.

Mechanical properties of cast films prepared from hydrogel

Cast films with a target thickness of $200 \pm 20 \mu\text{m}$ were prepared by pouring the hydrogel solution into a $12 \text{ cm} \times 12 \text{ cm}$ acrylic mold and drying under ambient conditions for 5 days. The dried films were peeled from the mold and cut into dog-bone specimens measuring 5 mm wide and 75 mm long. Film thickness was measured at five different locations, and five specimens were tested for each formulation. Before tensile testing, the films were conditioned for 24 h at 23°C and 50% relative humidity.

Tensile testing was performed using an INSTRON 5942 tensile tester (USA) equipped with Bluehill Universal software and a 500 N load cell, at a crosshead speed of 1 inch/min. The tensile testing workflow has been described in our previous work [17].

Biological properties of 3D bioprinted scaffolds

Hemocompatibility testing of 3D printed scaffolds Fresh human blood (6 mL) was collected from a healthy volunteer at the Medical Centre, BITS Pilani, K.K. Birla Goa Campus, after obtaining Human Ethical Clearance Certificate approval (Ref. no. HEC/BPCG/2023/007) and written informed consent under medical supervision. Of the collected blood, 4 mL was diluted with 5 mL of normal saline (0.85% sodium chloride). For the positive control, 0.5 mL of diluted blood was mixed with 9.5 mL of normal saline containing a small amount of sodium carbonate to induce hemolysis. In contrast, the negative control was prepared by mixing 0.5 mL of diluted blood with 9.5 mL of normal saline. For the test sample, the 3D-printed scaffold was immersed in 0.5 mL of diluted

blood mixed with 9.5 mL of normal saline. The positive control, negative control, and scaffold-containing test sample were incubated at 37 °C for 1 h in a water bath (EPS, SWB-02). After incubation, the tubes were centrifuged at 2000 rpm for 10 min at room temperature using a clinical centrifuge (Remi, C-24). The absorbance of the supernatant was measured at 545 nm using a UV–visible spectrophotometer (Shimadzu UV-1800), and percentage hemolysis was calculated using the Eq. (3) reported previously [18, 19].

$$\% \text{Hemolysis} = \left(\frac{\text{Absorbance of test sample} - \text{Absorbance of negative control}}{\text{Absorbance of positive control} - \text{Absorbance of negative control}} \right) \times 100 \quad (3)$$

Cell growth imaging via confocal microscopy Cell growth on crosslinked hydrogel was assessed using the *L929*, *mouse fibroblast* and *HaCaT*, *human keratinocyte cell* lines. A total of 1 mL of hydrogel was placed in a 35 mm Petri dish and crosslinked with 5% CaCl_2 . Prior to cell seeding, the hydrogels were sterilized under ultra-violet light. Cells were seeded at a density of 0.3×10^6 cells per sample, with cells-only coverslips serving as positive controls and hydrogels without cells as negative controls.

Samples were incubated for 24 and 48 h at 37 °C in a humidified atmosphere with 5% CO_2 . Following incubation, samples were fixed with 4% paraformaldehyde for 10 min at room temperature, washed with 1× PBS, and permeabilized with 0.1% Triton X-100 for 5 min. Samples were stained with Acti-stain 555 fluorescent phalloidin for 30 min in the dark, followed by washing with 1× PBS and counterstaining with Fluoroshield™ containing DAPI. Fluorescence images were acquired using a confocal microscope (Olympus FV3000; Olympus Corporation) at 20× magnification.

In-vitro cytotoxicity (MTT Assay) testing of hydrogel For quantitative analysis, the hydrogel sample was sterilized by autoclaving at 120 °C for 20 min and cross-linked with 5% w/v calcium chloride solution. The cross-linked hydrogel was then extracted in 1× PBS (phosphate-buffered saline) to obtain test solutions at concentrations of 10 and 20 mg/mL. *L929 mouse fibroblasts* and *HaCaT human keratinocytes* were seeded at a density of 0.05×10^6 cells per well in complete medium (DMEM supplemented with 10% FBS and 0.1% antibiotic-antimycotic) and incubated at 37 °C in a humidified atmosphere containing 5% CO_2 for 24 and 48 h. Cells treated with 1× PBS served as the control.

Following incubation, the culture medium was removed, and 300 μL of MTT reagent (3 mg/mL) was added to each well. The plates were incubated for 4 h to allow the formation of formazan crystals. Subsequently, the crystals were solubilized in 300 μL of DMSO and incubated for 5–10 min. The resulting solution was transferred to a 96-well plate, and absorbance was measured at 570 nm using a Thermo Scientific Multiskan GO microplate spectrophotometer. Cell viability was calculated from the absorbance values using the following Eq. (4) [20, 21].

$$\% \text{ Cell Viability} = \left(\frac{\text{Absorbance of test}}{\text{Absorbance of control}} \right) \times 100 \quad (4)$$

For qualitative assessment, representative micrographs of the control and hydrogel extract samples were acquired after incubation using an Olympus CKX53 inverted trinocular microscope at 10× magnification. Cellular morphology and overall growth characteristics were qualitatively assessed to evaluate cytotoxicity and in vitro biocompatibility.

Materials and preparation of drug-loaded bioink for drug delivery applications

The drug-loaded bioink was prepared using glimepiride as the active pharmaceutical ingredient, with Starch 1500® and maltodextrin as diluents, sodium alginate as the gelling agent, aspartame as the sweetener, and ponceau 4R aluminum lake as the colorant. All excipients are verified against the Inactive Ingredients Database (IID) to ensure compliance with established maximum daily exposure (MDE) limits. The ingredients were accurately weighed to obtain a total powder weight of 20 g, then dispersed in 113.3 mL of purified water. The mixture was stirred using a high-speed overhead stirrer for 90 min until a homogeneous bioink was obtained. Finally, approximately 5.1 mL of 5% w/v calcium chloride solution was added dropwise to form a semisolid printable hydrogel. The complete formulation is shown in Table 3.

3D-printing of glimepiride chewable tablets

A semisolid extrusion-based 3D bioprinting method was employed to produce the tablets. A Bioform bioprinter (Sytolab, India) equipped with a 10 mL Luer-Lock syringe was used for printing. The tablet designs were created using computer-aided design (CAD) software and the SuperSlicer software. The printing was conducted under ambient conditions, with each tablet weighing 1000 mg. The printing process and machine setup are mentioned in Table 4.

Printed tablets were frozen in a deep freezer (Esco) at -80 °C for 24 h, then lyophilized (freeze-dried) in a lyophilizer (Labconco) at -80 °C for an additional 24 h.

Table 3 Drug-loaded bioink formulation for glimepiride 2 mg tablets

Ingredients	Role	% w/w	mg/tablets
Glimepiride	Active Pharmaceutical Ingredient	1.33	2.00
Maltodextrin	Diluent	58.66	88.00
Starch 1500®	Diluent	17.00	25.50
Aspartame	Sweetener	2.00	3.00
Ponceau 4R Aluminum Lake	Colorant	1.00	1.50
Sodium alginate	Gelling Agent	20.00	30.00
	Total	100	150
Purified water	Media	15% w/v of hydrogel (113.3 mL water for 20 g powdered materials)	
Calcium chloride solution (5% w/v)	Crosslinker	5.1 mL crosslinker was added dropwise	
		100.00	150.00

Table 4 Printing setup and process parameters for glimepiride 2 mg tablets in a 3D bioprinter

3D printing machine description	Bioform Bioprinter, Kreator 3D, India
Printhead Z movement mechanism	Screw-based extrusion
Slicer software	Super slicer
Printing base	90 mm petri plate
Syringe	10 mL Luer-lock
Dispensing needle	Tapered tips 22 gauge (0.4 mm), Luer-lock
Dimensions (Xx Yx Z), mm	12×12×5.5
Shape	Round
Speed, mm/s	60
Extruder temperature, °C	Ambient
Bed temperature, °C	Ambient
Infill, %	90
Flow, %	150
Layer height, mm	0.55
No. of layers	11
Pattern	Line
Target printed tablet weight, mg	1000
Target lyophilized tablet weight, mg	150

The vacuum in the lyophilizer was set to 0.08 mbar. Once the freeze-drying process was complete, the lyophilized tablets were evaluated for various physical parameters, including average weight, dimensions, and hardness, as well as performance tests such as dissolution and content uniformity.

Physical parameters of 3D printed glimepiride 2 mg chewable tablets

The physical parameters of 3D-printed glimepiride tablets were evaluated through a series of tests to ensure their quality. First, the weight variation of twenty tablet samples was measured using an analytical balance (Make:

Sartorius, Model: Secura 513-101 M). Next, the average hardness of six tablets was measured using a semi-automatic tablet hardness tester (Pharmatron, MultiTest 50). Additionally, the dimensions of ten tablets, including thickness and diameter, were measured using a digital caliper (Make: BAKER, Model: SDN10). The mean results for all parameters were calculated and reported.

Dissolution testing of 3D printed glimepiride 2 mg chewable tablets

The release profile of 3D-printed glimepiride 2 mg tablets was evaluated using a tablet dissolution tester (Make: Electrolab, Model: EDT-08LX) equipped with apparatus II (paddle method). The reference dissolution method was adapted from the USP (United States Pharmacopoeia) Dissolution Testing, as described in the glimepiride tablets monograph [22]. The test was conducted at 75 rpm using 900 ml of phosphate buffer at pH 7.8, maintained at 37.0 ± 0.5 °C. The nine sampling time points were 15, 30, 45, 60, 90, 120, 150, 180, and 240 min.

Approximately 22 mg of glimepiride was weighed and placed in a 100 mL volumetric flask for standard preparation. Methanol was added, and the mixture was sonicated. The volume was then adjusted with methanol, diluted to 2 mL in 200 mL of the dissolution medium (pH 7.8) to achieve a concentration of 2.2 ppm.

One tablet was weighed and placed into each dissolution bowl (also at a concentration of 2.2 ppm) for sample preparation. After the designated time points, the collected samples were filtered through a 0.45 µm PVDF (polyvinylidene difluoride) filter, and the filtrate was transferred into HPLC (High Performance Liquid Chromatography) vials.

Glimepiride content was quantified using an Agilent 1260 series HPLC system with a C18 column (150×4.6 mm) at 35 °C, flow rate 1.5 mL/min, and injection volume 100 µL. Detection was performed at 228 nm, ensuring accurate results.

Content uniformity of 3D printed glimepiride 2 mg chewable tablets

The content uniformity test ensures that each tablet holds the proper amount of the drug. A standard solution was prepared by accurately weighing 20 mg of glimepiride and placing it in a 100 mL volumetric flask. A diluent was added, and the solution was sonicated until the glimepiride was fully dissolved. The volume was brought up to the mark with more diluent, resulting in a concentration of 20 ppm.

Five individual tablets were placed in separate 100 mL volumetric flasks for the sample preparation. A diluent of acetonitrile and water in a 90:10 ratio was added to fill approximately 70% of the flask's capacity. The mixture was sonicated to ensure complete dissolution,

after which the final volume was adjusted with the diluent to achieve a concentration of 20 ppm. The solution was filtered through a 0.45 μm polyvinylidene difluoride (PVDF) filter, and the first 3 mL of filtrate was discarded. The remaining solution was then transferred to HPLC vials for analysis.

The drug content of each of the five individual 3D-printed glimepiride 2 mg chewable tablets was determined by high-performance liquid chromatography (HPLC). For the analysis, glimepiride was quantified at 228 nm using an Agilent 1260 series HPLC system. Chromatographic separation was achieved using a C18 column (150 $\times$ 4.6 mm) maintained at 35 $^{\circ}\text{C}$. The mobile phase was delivered at a flow rate of 1 mL/min, with an injection volume set at 20 μL . Detection at 228 nm provided accurate and reliable quantification of glimepiride content in each tablet sample.

Results and discussion

Materials and preparation of hydrogel for tissue engineering applications

Pharmaceutical polymers commonly used in tablet formulations were selected for hydrogel preparation because of their established regulatory acceptance, consistent quality, wide availability, cost-effectiveness, non-toxicity, biodegradability, and biocompatibility.

Starch 1500 $^{\circ}$, a partially pregelatinized modified starch, was included for its multifunctional pharmaceutical properties and ability to hydrate in cold water, thereby increasing viscosity and facilitating hydrogel processing. Starch-based scaffolds have previously been investigated for bone tissue engineering applications [23, 24], and starch-containing bioinks have been reported to support three-dimensional cell growth [25]. Maltodextrin (Glucidex 12D), commonly used as a filler in chewable tablets, was incorporated to improve printability, mechanical performance, and osmolarity regulation. Previous studies have also indicated its potential for 3D-printed bone and skin tissue engineering, where it can support fibroblast proliferation [26, 27]. Sodium alginate was selected as the gelling polymer because of its water solubility, shear-thinning behavior, rapid ionic gelation with multivalent cations such as Ca^{2+} , biocompatibility, and ability to support cell growth [28]. Together, Starch 1500 $^{\circ}$, maltodextrin, and sodium alginate produced an animal-free, homogeneous, lump-free hydrogel that was easy to prepare. The key hydrogel properties are summarized in Table 5.

Physicochemical properties of hydrogel and 3D bioprinted scaffolds

Printability assessment results of hydrogel The printability of the crosslinked hydrogel was proved by the successful fabrication of multiple scaffold geometries under

ambient conditions using semisolid extrusion-based 3D bioprinting (Fig. 1). As shown in Fig. 1A–C, the hydrogel yielded a well-defined single-layer 10 mm $\times$ 10 mm square grid scaffold with uniform filament deposition, with an average filament diameter of 0.90 ± 0.06 mm and a pore geometry value of 1.04 ± 0.07 with a perfect square shape, indicating good printing precision and shape fidelity. In addition, the hydrogel enabled fabrication of a 15 mm spiral construct (Fig. 1D) and a five-layer 30 mm $\times$ 30 mm square grid scaffold that preserved its architecture in both top and side views (Fig. 1E–F), confirming its ability to support shape versatility and layer-by-layer stacking. These observations are consistent with the rheological behavior of the hydrogel, including high viscosity, shear-thinning behavior, thixotropic recovery, and predominantly elastic behavior at rest, which collectively favor smooth extrusion and structural stability after deposition. The overall printability assessment is further summarized in Table 5.

The study shows that the crosslinked hydrogel used in the printing assessment produces consistent filaments while preserving their pore geometry, resulting in a nearly perfect square shape. This leads to high printability ($\text{Pr} = 1$) without clogging, ensuring that the scaffold structure remains intact [29]. These findings highlight the hydrogel's effectiveness for bioprinting, making it suitable for use with cells and therapeutic agents.

Viscosity results of hydrogel Viscosity is a key determinant of hydrogel and bioink printability in extrusion-based 3D printing. Highly viscous formulations can improve post-extrusion shape retention. Still, they may increase the risk of nozzle blockage, whereas low-viscosity systems are easier to extrude but may compromise filament stability and print resolution [30].

As summarized in Table 5, the crosslinked hydrogel exhibited a viscosity of 1.56×10^6 mPa·s, which falls within the reported printable range for extrusion-based 3D bioprinting (approximately 30 mPa·s to 6×10^7 mPa·s), depending on formulation composition and printing conditions [30].

Printing trials further confirmed that this viscosity supported reliable extrusion and satisfactory printability. The relatively high viscosity may also reduce drug sedimentation during printing, thereby improving content uniformity in low-dose formulations such as the 2 mg 3D-printed glimepiride tablets.

Rheological characteristics of crosslinked hydrogel

Rheological characterization is a critical requirement for assessing the suitability of hydrogels for extrusion-based bioprinting, as it offers fundamental insight into flow behavior during printing and structural recovery

Table 5 Hydrogel properties assessment results

Property	Unit	Results	Inference
"Preparation" of hydrogel		Smooth and uniform in 90 min	Quick and easy to prepare
Printability	Various shapes: Square and spiral Number of layers Filament uniformity Pore geometry	Satisfactory printability 5 layers 0.90 ± 0.06 mm 1.04 ± 0.07	Printable under ambient conditions Suitable for multilayer printing Precise printability Good shape fidelity
Viscosity	mPa.s	1.56 × 10 ⁻⁶ ± 0.1	Suitable for semisolid extrusion printing
Rheology	Shear thinning Thixotropy	Viscosity decreases upon applying shear stress. 87% recovery	Suitable for extrusion-based printing Good structural recovery, upon shear removal
	Viscoelasticity Yield stress Flow point	Elastic modulus is higher than the viscous modulus 20 Pa 108 Pa	Better shape fidelity upon printing Favorable for extrusion printing and shape retention
Swelling of scaffolds in media	% Swelling	72% in 24 h	Rapid swelling
In vitro degradation rate	% Mass loss	82% in 35 days	Suitable for skin tissue engineering
Surface morphology	% Porosity & Pore size (µm)	62.7 ± 0.1 39.2 ± 13.8	Suitable for skin tissue engineering
Mechanical properties of cast film	Tensile strength Extension at break	33.9 ± 3.4 MPa 4.5 ± 0.6%	High tensile strength Low extensible
Hemocompatibility	% Hemolysis	5% ± 0.0%	Highly hemocompatible
Cell imaging with skin cells	<i>L929 cells</i> <i>HaCaT cells</i>	Faster growth, Slow growth	Suitable for tissue skin engineering
<i>L929 and HaCaT cells</i>			
In-vitro cytotoxicity with skin cells	% Cell viability Microscopic images	• ≥ 70% • Consistent cell growth	Non-toxic and compatible with skin cells
<i>L929 and HaCaT cells</i>			

following deposition. In the present study, the cross-linked hydrogel was comprehensively evaluated for shear-thinning behaviour, thixotropic recovery, viscoelasticity, yield stress, and flow point. The corresponding results are presented in Figs. 2A and 3D, and the principal inferences drawn from these measurements are summarized in Table 5.

Shear thinning, or a continuous decline in viscosity with increasing shear rate, is a vital fluid behaviour for extrusion-based bioprinting. As shown in Fig. 2A, our crosslinked hydrogel sample exhibits a continuous decrease in viscosity with increasing shear rate, indicating its suitability for bioprinting [29].

Rapid viscosity recovery after extrusion is essential to prevent structural collapse and to help the printed filament retain its shape [31]. Figure 2B illustrates the thixotropic behaviour of the crosslinked hydrogel, a time-dependent shear-thinning property. When the shear rate was increased abruptly from 0.1 to 100 s⁻¹, the viscosity dropped immediately to a low value, facilitating extrusion. When the shear rate was reduced again to 0.1 s⁻¹, the viscosity recovered quickly to 87%, indicating good structural recovery after shear removal.

The viscoelastic behaviour of the crosslinked hydrogel was evaluated using frequency-sweep analysis, which reflects the balance between its elastic and viscous responses. As shown in Fig. 2C, the elastic modulus (G') remained higher than the viscous modulus (G'') across the entire angular frequency range, indicating that the hydrogel exhibited solid-like behaviour at rest. This viscoelastic profile is advantageous for extrusion-based 3D bioprinting because it suggests a stable internal network that can resist deformation after deposition. Consequently, the extruded filaments are better able to retain their geometry, support layer-by-layer stacking, and maintain overall shape fidelity during printing.

Yield stress and flow point are important rheological parameters for extrusion-based bioprinting because they define the stress needed to initiate flow and the transition to liquid-like behavior [31]. As shown in Fig. 2D, the crosslinked hydrogel exhibited a yield stress of 20 Pa and a flow point of 108 Pa, indicating favorable extrusion behavior with adequate post-printing shape retention.

% Swelling and invitro degradation rate of 3D bio-printed scaffolds The swelling and degradation rates of biomaterial scaffolds are critical factors in tissue engi-

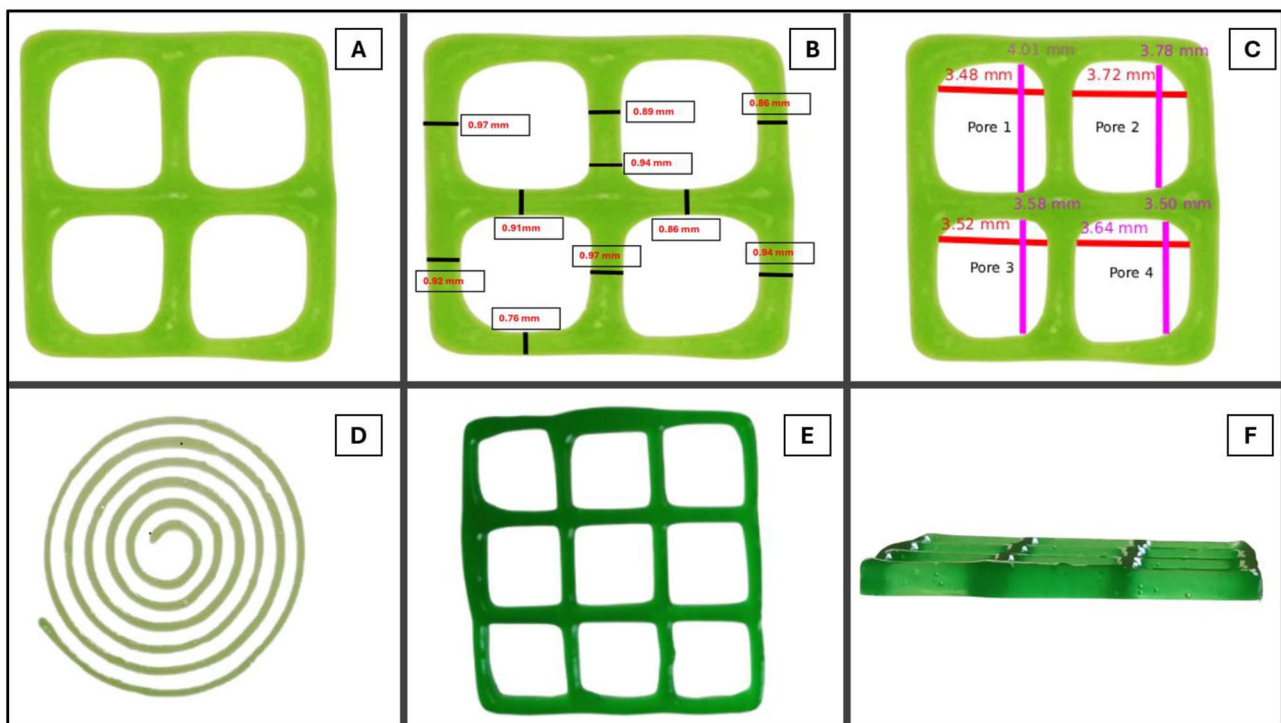


Fig. 1 Semisolid extrusion-based 3D biprinting of hydrogel: **(A)** One layered, Square-shaped 10 mm x 10 mm, grid scaffold **(B)** One layered, Square-shaped 10 mm x 10 mm grid scaffold with printed filament uniformity at 10 random points, average filament diameter- $0.90 \text{ mm} \pm 0.06$, **(C)** One layered, Square-shaped 10 mm x 10 mm grid scaffold with pore geometry assessment, average pore geometry- 1.04 ± 0.07 **(D)** One layered, Spiral-shaped scaffold with print 15 mm diameter. **(E)** Five-layered, Square-shaped 30 mm x 30 mm grid scaffold, Top view **(F)** Five-layered, Square-shaped 30 mm x 30 mm grid scaffold, side view

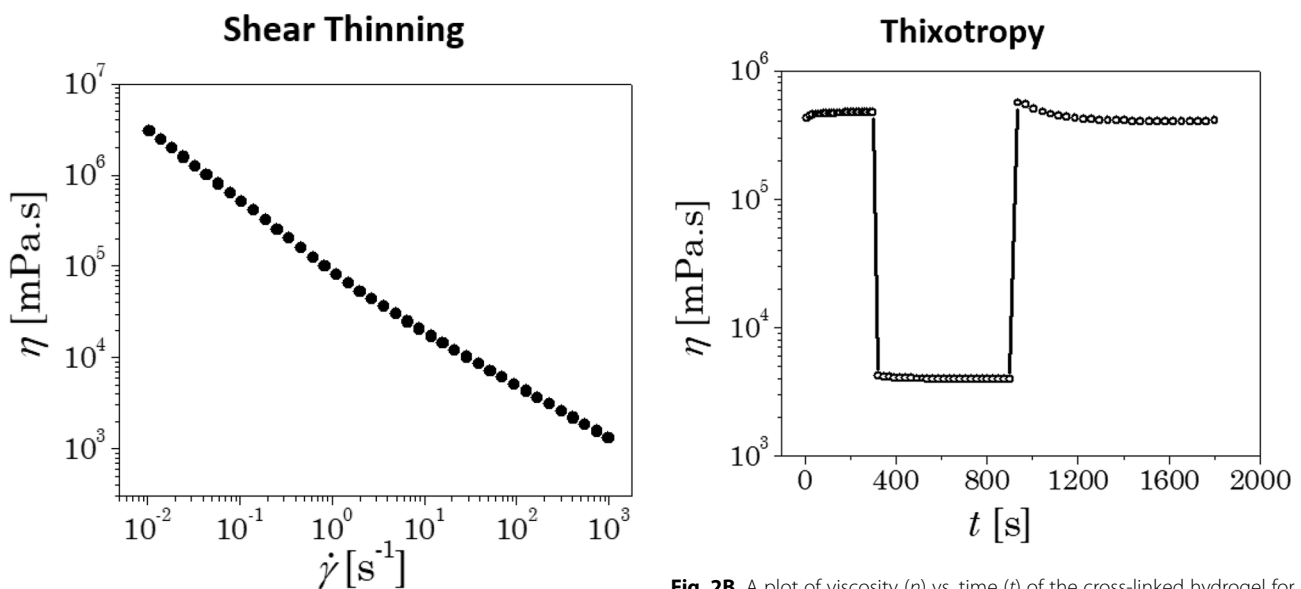


Fig. 2A A plot of viscosity (η) vs. shear rate ($\dot{\gamma}$) of the cross-linked hydrogel demonstrating shear thinning behaviour

Fig. 2B A plot of viscosity (η) vs. time (t) of the cross-linked hydrogel for the recovery test. The viscosity recovery percentage is 87%

neering applications. When scaffolds absorb media and swell, they create a hydrated environment that closely resembles the natural extracellular matrix. This hydration supports essential cellular processes such as cell attach-

ment, migration, and proliferation, all of which are necessary for successful tissue engineering, particularly in skin regeneration.

Optimal swelling is necessary to maintain a balance between mechanical stability and biological function.

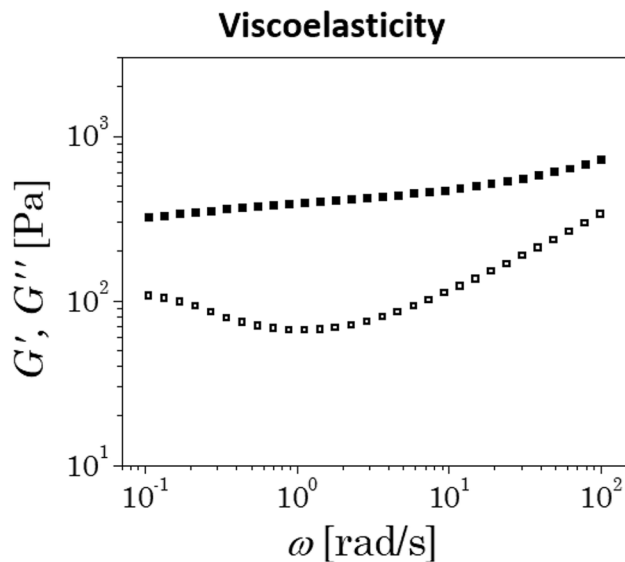


Fig. 2C The elastic modulus, G' (filled symbols), and viscous modulus, G'' (open symbols), of the cross-linked hydrogel as a function of angular frequency (ω)

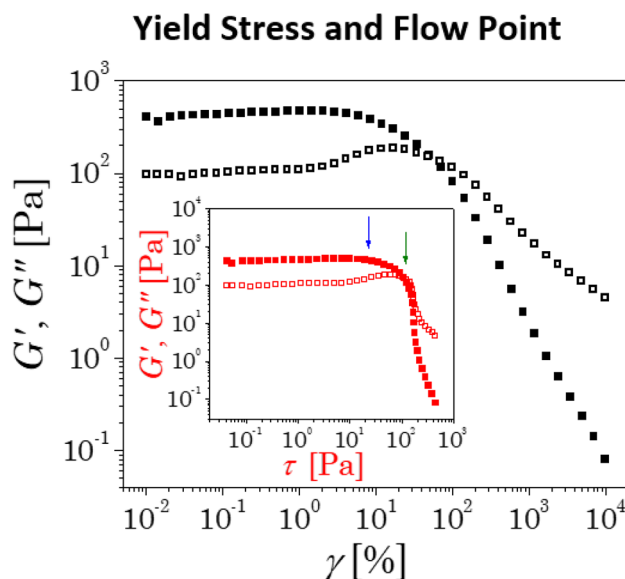


Fig. 2D The elastic modulus, G' (filled symbols) and viscous modulus, G'' (open symbols) of the cross-linked hydrogel as a function of strain amplitude, γ . The inset shows the G' and G'' data shown in the main figure plotted against stress amplitude τ . The yield stress, τ_y , is 20 Pa (marked with blue arrow) while the flow point, τ_f , is 108 Pa (marked with green arrow)

Excessive swelling can compromise the scaffold's structural integrity, while insufficient swelling may restrict cell growth and limit nutrient exchange. Achieving the right level of swelling ensures that the scaffold remains robust enough to support tissue formation while providing an environment conducive to cellular activity. In our experiments, the scaffolds showed rapid swelling, reaching up to 72% within the first 24 h (Fig. 3A).

For successful skin tissue engineering, supporting scaffold integrity during the early stages is crucial. Specifically, the scaffold should remain intact for the first three weeks, providing essential support for wound healing and facilitating the formation of new tissue formation [32]. In the present study, the 3D bioprinted scaffolds showed notable swelling behavior that persisted for up to two weeks. After this period, a gradual mass loss was observed, indicating the onset of in vitro degradation. This transition is considered beneficial for tissue regeneration, as it indicates that the scaffold structure is being replaced by newly forming tissue.

Following the initial phase, a more pronounced degradation was detected, with approximately 82% of the scaffolding material breaking down between days 17 and 35, as illustrated in Fig. 3B. This degradation profile is advantageous because it ensures the scaffold provides mechanical support during the critical early period and then allows for controlled, gradual replacement by regenerating tissue [32, 33]. The observations regarding swelling and degradation rates are summarized in Table 5.

Porosity of the scaffolds through FESEM Porosity is an important structural feature of tissue-engineered scaffolds because it supports cell migration, fluid transport, nutrient exchange, and tissue formation. In 3D bioprinting, porosity can be controlled by printing parameters such as infill percentage.

FESEM analysis (Fig. 4) revealed pore sizes of 20–74 μm , with a mean pore size of $39.2 \pm 13.8 \mu\text{m}$, confirming a porous scaffold architecture relevant to skin tissue engineering. Pore size is a key determinant of scaffold performance, as it influences cell attachment, migration, vascularization, and nutrient diffusion. Very small pores (1–2 μm) favor epidermal cell attachment, whereas pores of 2–12 μm support dermal cell migration. Pores in the 40–100 μm range promote vascular structure formation, while pores of 90–200 μm are considered optimal for cell infiltration and nutrient transport. Although pores larger than 200 μm may improve oxygen and nutrient exchange, they can also reduce cell density by promoting excessive fibroblast migration. Therefore, an optimum pore-size distribution is essential for effective skin regeneration [34–36].

ImageJ analysis revealed a porosity of $62.7 \pm 0.1\%$, which falls within the 60–90% range considered favourable for wound healing and skin tissue engineering, as it supports cellular activity, oxygen and nutrient diffusion, and extracellular matrix deposition [37].

Chemical analysis results of the scaffolds through FTIR Crosslinking is crucial for providing stability and strength to bioinks in 3D bioprinting. In our formulation, sodium alginate acts as the gelling agent, and cal-

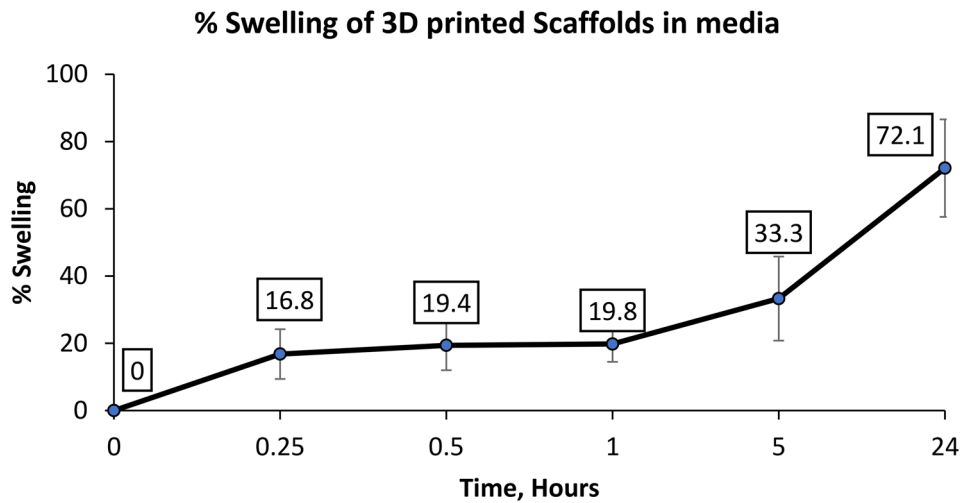


Fig. 3A % Swelling of 3D printed scaffolds over the period of 24 h

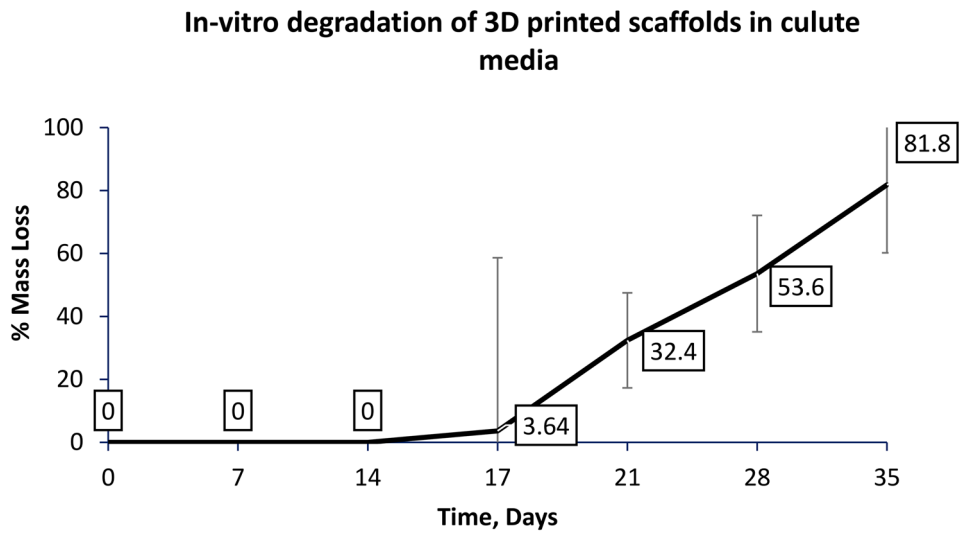


Fig. 3B % Mass loss as a representative of in vitro degradation of 3D printed scaffolds over a period of 35 days

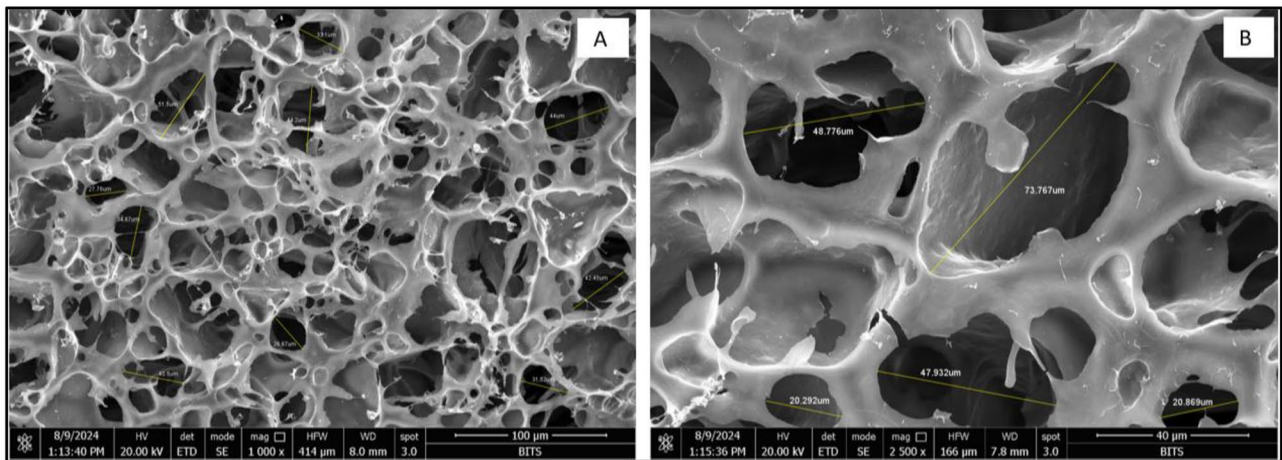


Fig. 4 Lyophilized scaffolds image under FESEM: (A) at 1000x magnification, (B) at 2500x magnification

cium chloride serves as the ionic crosslinker. FTIR spectroscopy of un-crosslinked and crosslinked hydrogels revealed a COO^- band shift (1596 cm^{-1} to 1638 cm^{-1}) and a clearer peak at 1411 cm^{-1} , confirming robust calcium-alginate “egg-box” junctions. The O–H stretch moved from 3273 cm^{-1} to 3308 cm^{-1} with decreased intensity, indicating reduced chain mobility. More defined C–O–C and C–O peaks ($1153, 1082, 1032\text{ cm}^{-1}$) suggested a more ordered structure. Figure 5 illustrates the relevant FTIR spectra and the effect of crosslinking.

Mechanical properties of cast films prepared from hydrogel

The cast films exhibited high tensile strength ($33.9 \pm 3.4\text{ MPa}$), which falls within the broad mechanical range reported for human skin (1.09 kPa to 83.3 MPa). In contrast, the extension at break was only $4.5 \pm 0.6\%$, indicating low extensibility compared with the elongation values typically considered desirable for skin scaffolds

(35% to 120%) [37]. This mechanical profile suggests that the material can provide good structural support. Still, it may be less suitable for highly mobile wound sites, where low flexibility could increase the risk of cracking or detachment. Accordingly, in its current form, the formulation may be better suited to low-mobility anatomical regions where mechanical stability is more critical than stretchability.

A limitation of this analysis is that mechanical testing was performed on cast films rather than directly on 3D-printed scaffolds. Cast films were used to obtain uniform, reproducible material-level properties; however, the mechanical behaviour of printed scaffolds may differ due to structural features such as porosity, filament architecture, and interlayer interfaces. Therefore, these results should be interpreted as indicative of the material's intrinsic properties rather than the exact mechanical performance of the final printed constructs.

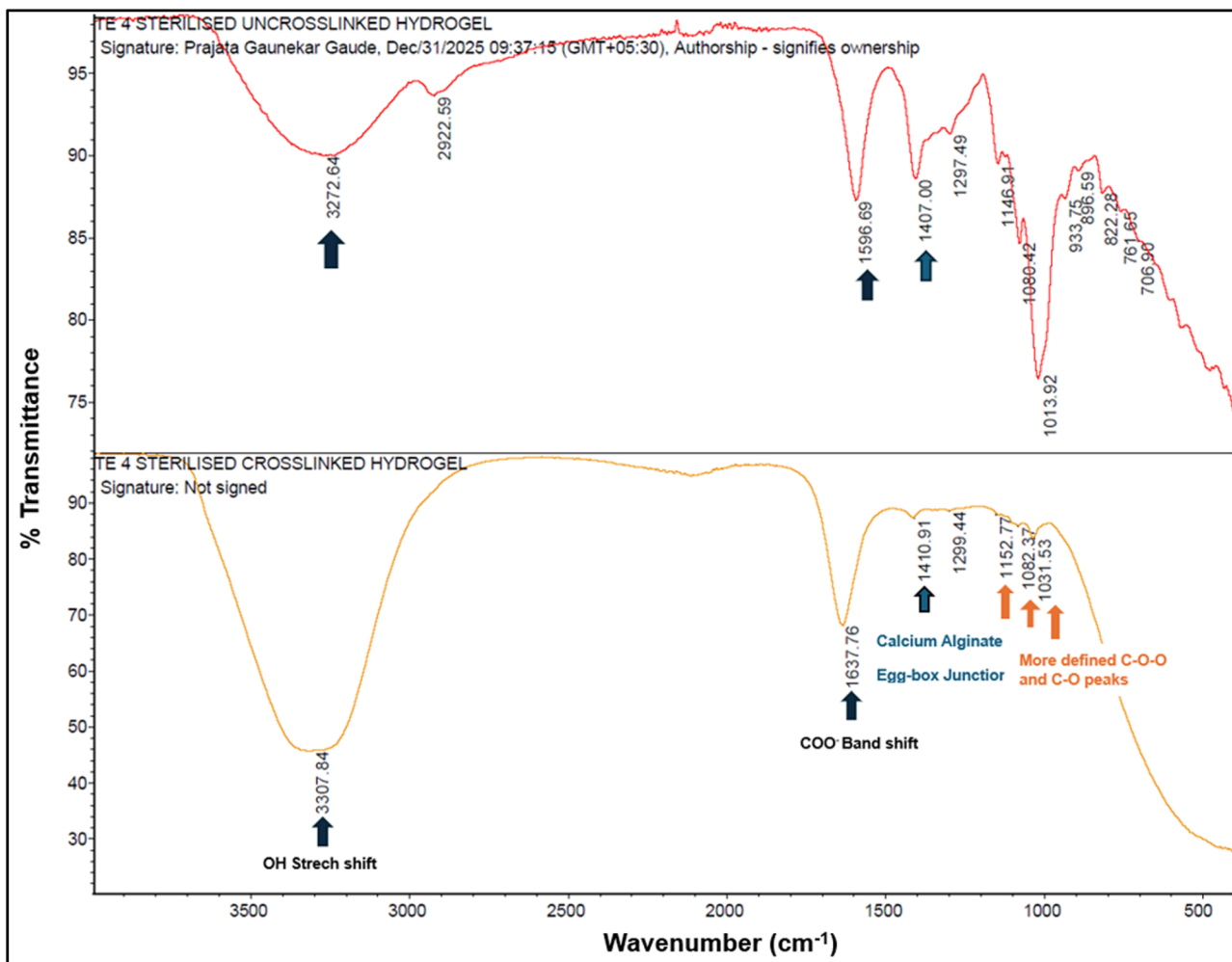


Fig. 5 FTIR spectra comparing un-crosslinked and crosslinked hydrogels. Arrows indicate peak shifts and changes resulting from crosslinking

Biological properties of 3D bioprinted scaffolds

Hemocompatibility testing results of 3D printed scaffolds Hemocompatibility is an essential safety assessment conducted on implants, medical devices, and long-acting drug delivery systems to evaluate their compatibility with blood. Hemocompatibility tests can be performed in vitro, ex vivo, and in vivo [38].

In our study, we performed in vitro hemocompatibility test on freeze-dried hydrogel scaffolds. The results showed that the samples were hemocompatible, with only $5 \pm 0.0\%$ hemolysis. Hemocompatibility can be categorized as follows: highly hemocompatible ($< 5\%$ hemolysis), hemocompatible ($5\text{--}10\%$ hemolysis), and non-hemocompatible ($> 20\%$ hemolysis) [18].

Figure 6 illustrates the percentage of hemolysis for the 3D printed scaffold, and the findings are summarized in Table 5.

Cell growth imaging via confocal microscopy Cell imaging is essential in tissue engineering for assessing cell viability and understanding intercellular interactions and organization. Confocal microscopy is particularly valuable, providing high-resolution, three-dimensional visualization of cell behavior on scaffolds.

Our confocal imaging demonstrates the successful integration and proliferation of *L929 mouse fibroblast cells*, along with the gradual growth of *HaCaT human keratinocytes* within crosslinked hydrogels. At 48 h, cell growth surpassed that at 24 h, particularly in *L929 cells*, which showed consistent growth with clearly visible nuclei and cytoskeletons. In contrast, *HaCaT cells* grew more slowly,

likely because Ca^{2+} in the hydrogel served as a crosslinker [39]. These findings highlight the efficacy of the hydrogels in promoting sustained cell expansion, confirming their suitability for skin tissue engineering. This is illustrated in Fig. 7A and B, and the observations are further captured in Table 5.

In vitro cytotoxicity (MTT Assay) results of the hydrogel

In vivo cytotoxicity testing is a key approach for evaluating the biocompatibility of biomaterials. Among the available methods, the MTT assay is widely used to assess metabolic activity, cytotoxicity, and cell viability [40]. In the present study, hydrogel extracts prepared in $1\times$ PBS showed no detectable cytotoxicity and satisfied the accepted biocompatibility criterion, with cell viability remaining above 70% [20].

The cytotoxicity of the hydrogel extracts was evaluated over 48 h using the *L929 mouse fibroblast* and *HaCaT human keratinocyte cell* lines. In *L929 cells*, viability ranged from 75% to 90% across all tested conditions, indicating non-cytotoxicity and good cytocompatibility. In *HaCaT cells*, viability at 24 h ranged from 55% to 66%; however, it increased to $>80\%$ at 48 h. Overall, these findings indicate that the hydrogel extracts were non-cytotoxic and biocompatible at the tested concentrations.

Qualitative observations supported the quantitative findings. *L929 cells* retained healthy spindle-shaped morphology and sustained growth. In contrast, *HaCaT cells* remained viable. The cells showed slow growth with a flat, polygonal shape and increased in number from 24 to 48 h at all tested concentrations. This indicates that the

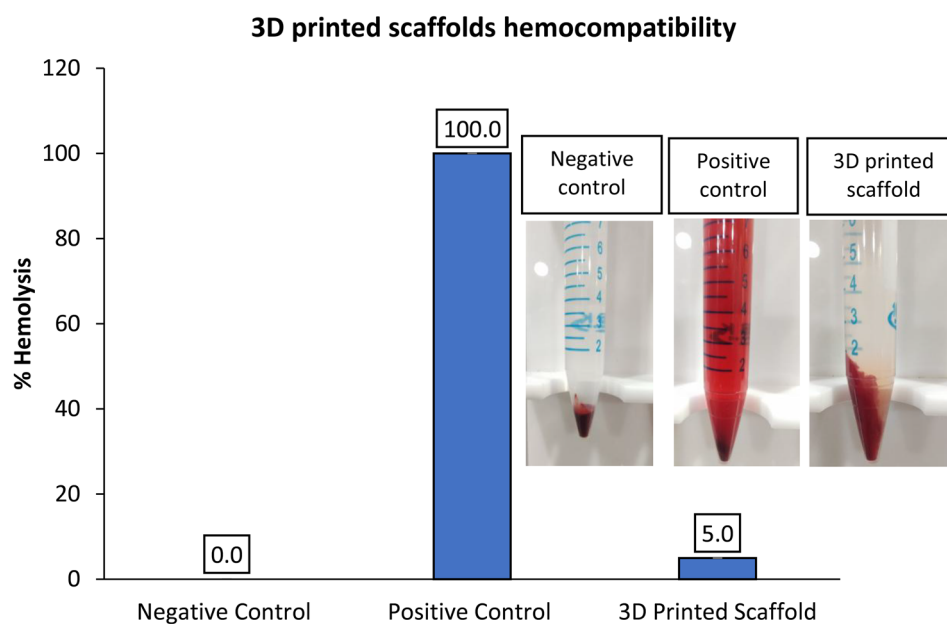


Fig. 6 Graphical representation of the % hemolysis of the 3D printed scaffold sample

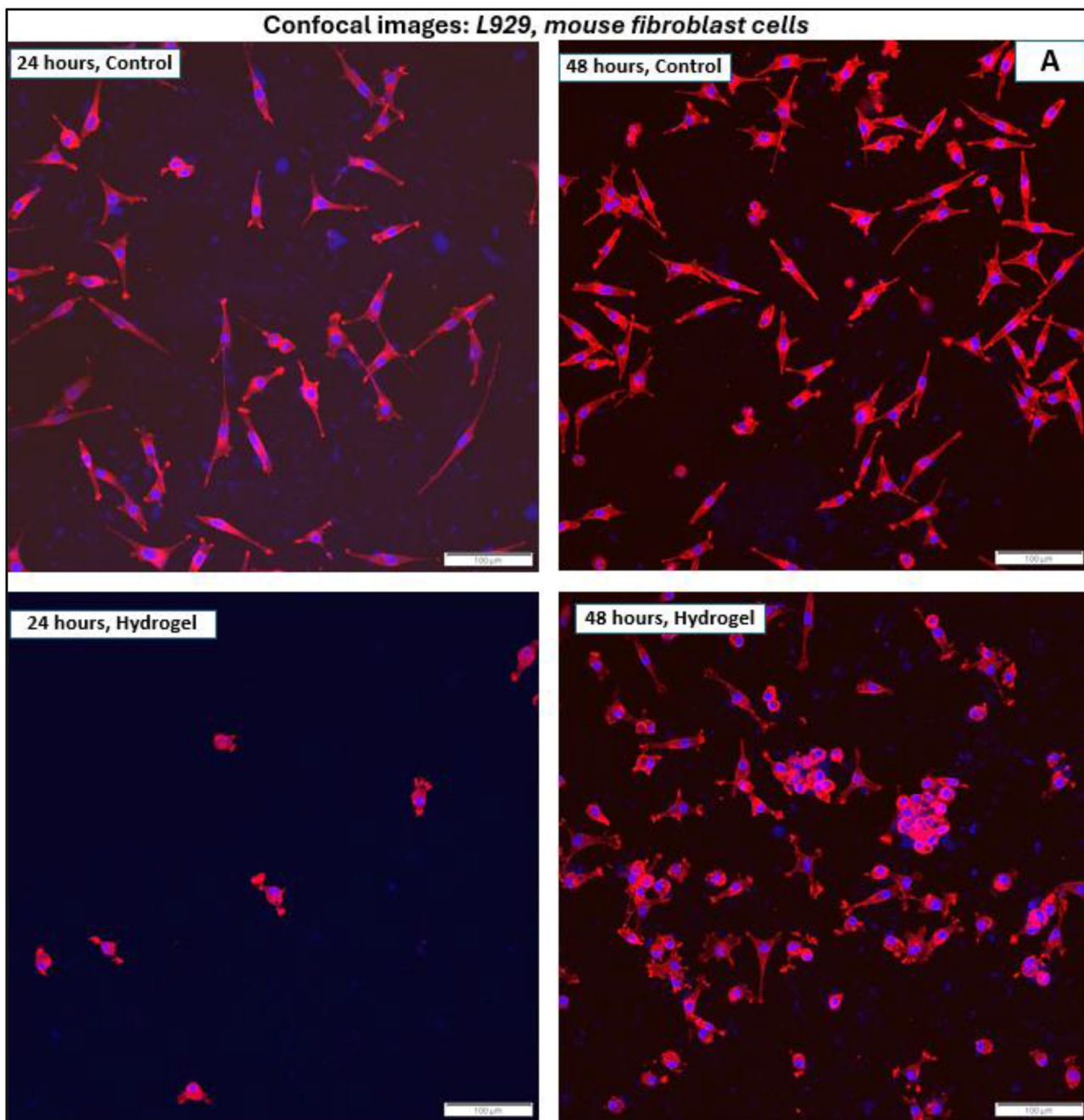


Fig. 7A Confocal microscopy of *L929, mouse fibroblast cells* cultured on crosslinked hydrogel demonstrates intact nuclei (DAPI, blue) and well-organized cytoskeleton (Phalloidin, red), with enhanced cell growth observed at 48 h. Magnification = 20x, Scale = 100 µm

hydrogel extracts were safe and promoted cell growth in skin-related cell lines.

The high cell viability in both cell lines supports the suitability of the hydrogel for skin tissue engineering and 3D bioprinting applications. The corresponding data are shown in Fig. 8 (A-D) and summarized in Table 5.

Materials and preparation of drug-loaded bioink for drug delivery applications

For our 3D printed drug delivery application of low-dose glimepiride bioink formulation, we selected Starch[®] and maltodextrin as diluents to provide necessary bulk and support the structural integrity of the tablets. Sodium alginate was incorporated to impart optimal gelling characteristics, ensuring the bioink achieved the viscosity required for smooth semisolid extrusion during 3D

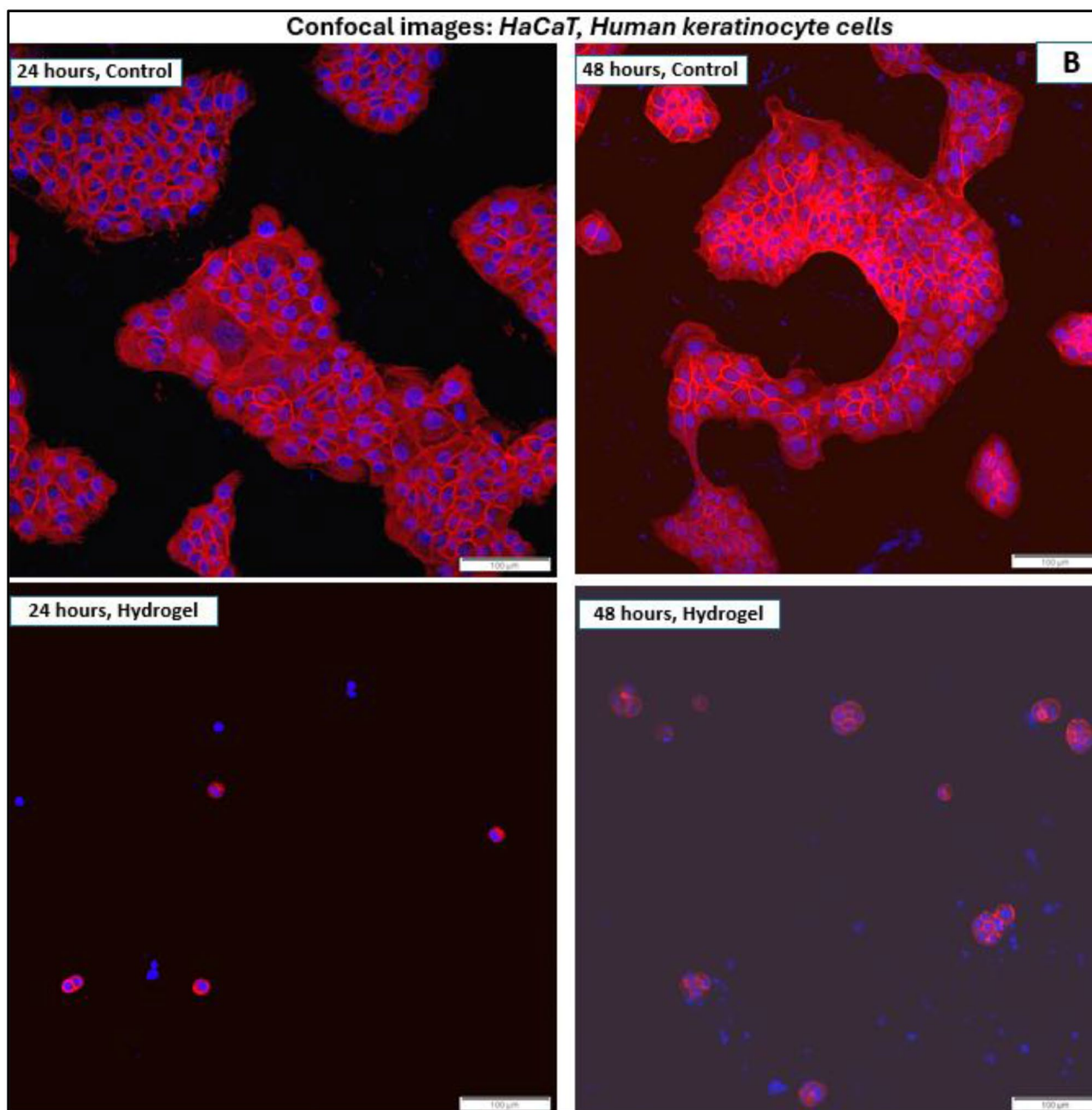


Fig. 7B Confocal microscopy of *HaCaT*, human keratinocyte cells cultured on crosslinked hydrogel, shows gradual cell growth. Nuclei are stained with DAPI (blue) and the cytoskeleton with Phalloidin (red). Slow cell growth is observed at 20x magnification; scale bar = 100 µm

printing. To enhance patient experience and compliance, we also added a colorant for visual appeal and a sweetener to deliver a pleasant taste, recognizing that most chewable tablets benefit from a sweet flavor profile. The resulting bioink mixture was homogeneous and lump-free, with a viscosity that enabled reliable, consistent 3D printing performance.

3D-printing of glimepiride 2 mg chewable tablets

Glimepiride tablets were fabricated using a Bioform bio-printer to enable reproducible semisolid extrusion-based 3D printing. Each tablet was printed to an initial target weight of approximately 1000 mg to provide sufficient bulk and structural integrity for subsequent processing. The printed tablets were then frozen to preserve their geometry before drying.

Following freezing, the tablets were lyophilized to remove moisture while maintaining their overall shape

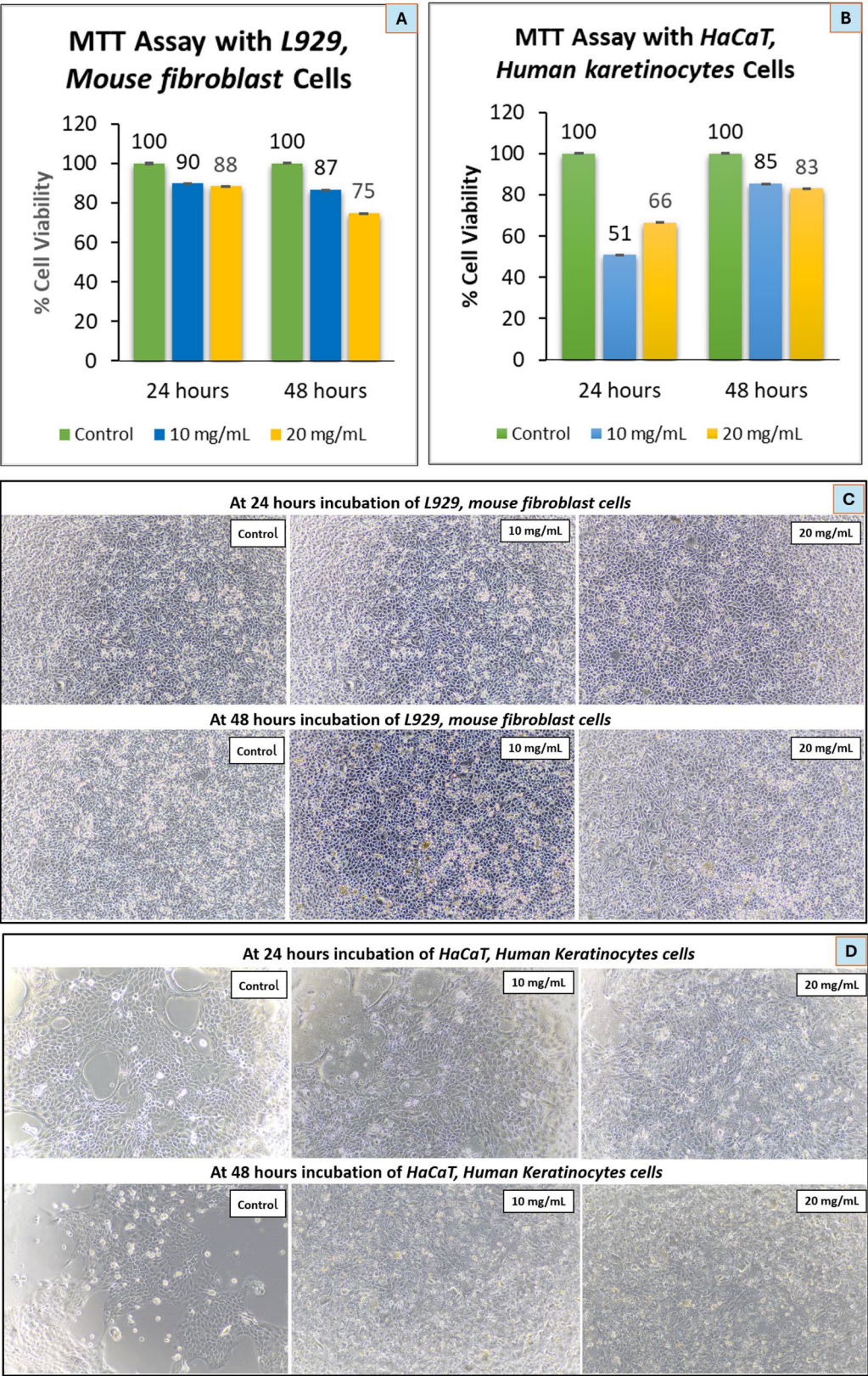
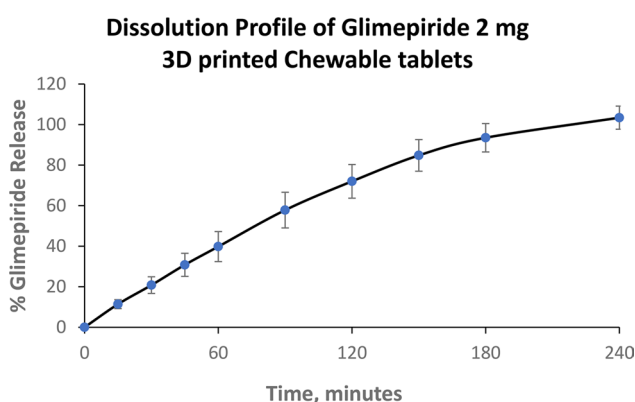


Table 6 Physical properties of glimepiride 2 mg 3D printed chewable tablets

Property	Unit	Results	Inference
Tablet weight variation	Milligrams (mg)	154.9 ± 7.4	Consistent tablet weight
Average tablet hardness	Kilopond (kp)	19.7 ± 7.4	Hard tablets
Average tablet diameter	Millimetres (mm)	13.7 ± 0.7	Acceptable
Average tablet thickness	Millimetres (mm)	6.3 ± 0.7	Acceptable

**Fig. 9** Dissolution profile of glimepiride 2 mg 3D printed chewable tablets

and physical integrity. The resulting tablets weighed approximately 150 mg, were round, and showed an acceptable physical appearance, indicating successful formulation and processing.

Printing occurred without nozzle clogging, confirming the bioink's suitability for semisolid extrusion. Ambient printing followed by lyophilization also avoided heat exposure, making the process suitable for heat-sensitive drugs.

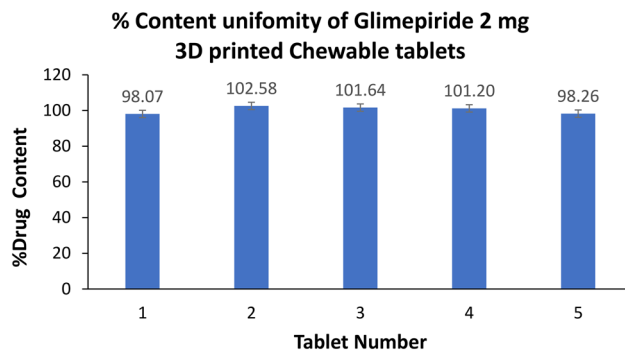
Physical parameters of 3D printed glimepiride 2 mg Tablets

Tablet physical properties, including weight, hardness, shape, and size, are key quality attributes that influence manufacturing consistency, dosing accuracy, and patient acceptability.

As shown in Table 6, the 3D-printed glimepiride 2 mg tablets showed controlled weight variation within 7.5% of the mean tablet weight of 150 mg, indicating good printing precision and dose consistency.

Tablet hardness supported mechanical integrity and reduced the risk of breakage during handling, packaging, and transport, indicating robust tablets produced by semisolid extrusion-based 3D printing.

The round chewable design, combined with sweetener and colourant, may improve ease of administration, palatability, visual appeal, and patient adherence,

**Fig. 10** Content uniformity test results for glimepiride 2 mg 3D printed chewable tablets

particularly for individuals who have difficulty swallowing conventional tablets.

Overall, the 3D-printed glimepiride tablets exhibited consistent weight, high hardness, and acceptable dimensions, indicating good printing precision, mechanical robustness, and suitability as chewable tablets.

Dissolution results of glimepiride 2 mg 3D printed chewable tablets

Dissolution testing is a critical In-vitro assessment for solid oral dosage forms, providing information on formulation performance, product comparability, potential in vivo behaviour, and batch-to-batch consistency.

The 3D-printed glimepiride 2 mg tablets exhibited drug release over 4 h (Fig. 9), indicating an extended-release profile. This behaviour may be attributed to the poor aqueous solubility of glimepiride and its entrapment within the crosslinked hydrogel matrix formed during semisolid extrusion printing, which likely restricted drug diffusion. As no pharmacopeial specification or reference-product comparison was included, the dissolution study was intended to characterize the release behaviour of this prototype formulation and assess whether it more closely resembled an immediate- or extended-release system.

Extended-release glimepiride may improve therapeutic consistency, reduce dosing frequency, and enhance adherence through gradual drug release.

Content uniformity of tablets

Content uniformity is a critical quality attribute for low-dose formulations. According to USP General Chapter <905>, uniformity of dosage units is required for formulations containing less than 25 mg of drug or in which the drug substance constitutes less than 25% of the dosage unit. In conventional low-dose tablets, achieving uniform drug distribution can be difficult because active pharmaceutical ingredients may segregate during direct compression; wet granulation is therefore commonly

used to reduce this risk, although it adds processing complexity and resource demand.

In this study, the 3D-printed glimepiride 2 mg tablets showed excellent content uniformity, with a mean drug content of 100.4% and a standard deviation of 2.8% across five tablets (Fig. 10). These findings indicate that semisolid extrusion-based 3D printing can produce low-dose tablets with consistent dosage units. The high viscosity of the crosslinked hydrogel likely helped maintain glimepiride particles in suspension during printing, thereby limiting sedimentation and supporting uniform drug distribution in the final tablets.

Conclusion

This study demonstrates the potential of a pharmaceutical polymer-based hydrogel composed of Starch 1500®, maltodextrin and sodium alginate as a bioink for semisolid extrusion-based 3D bioprinting.

The crosslinked hydrogel showed suitable printability, viscosity and rheological behaviour. The printed scaffolds exhibited favourable swelling–degradation characteristics, pore architecture, porosity, hemocompatibility, non-cytotoxicity and support for the growth of *L929* and *HaCaT* cells, indicating their potential for skin tissue engineering. Cast films showed tensile strengths within the reported range for human skin. Since mechanical testing was performed on cast films instead of printed scaffolds, the results reflect material-level properties and may not fully reflect the mechanical performance of the final constructs.

The 3D-printed glimepiride tablets showed acceptable physical properties, consistent drug content and controlled release over 4 h, supporting their potential for precise low-dose oral drug delivery.

Overall, these findings demonstrate the versatility of pharmaceutical polymer-based hydrogels as scalable, regulatory-aligned bioink platforms for tissue engineering and personalized oral drug delivery.

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Author contributions

H. K. B.- Conceptualization, investigations, performed drug delivery experiments, hydrogel testing's and wrote the original manuscript draft. M. S.- Manuscript review and editing. A. P. R. Execution of confocal imaging and in vivo cyto-toxicity experiments. P. P. B.- Execution of hemocompatibility experiment. S. B. K.- Execution of rheological experiment (Revised testing-Shear thinning, yield stress, thixotropy and viscoelasticity). M. K. T.- Execution of rheological experiment (Shear thinning). A. S.- Supervision, manuscript drafting and review of rheological experiment. A. G.- Project administration, supervision, resource allocation and review of manuscript.

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Data availability

All data supporting the findings of this study are available within the manuscript and additional raw data will be available from the corresponding author upon reasonable request.

Declarations

Ethical approval

This study did not include any animal experiments. We used blood from healthy human volunteers and obtained ethics committee approval (certificate number HEC/BPCG/2023/007) for the hemocompatibility study, which adhered to Protocol no. HEC/BPCG/2003/007.

Competing interests

The authors declare no competing interests.

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